INTEGRAL UNIVERSALLY BIJECTIVE MORPHISMS PRESERVE AND REFLECT SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3724-3735, following the statement at 3717-3722.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Open subspaces. Properties of Spaces, Lemma spaces-properties-lemma-open-subspaces (cross-chapter, cited at ~3729-3731). V is the open subspace of Y with |V| = |f|(|U|).
- Integral surjective target is affine. proposition-affine (same chapter, card L26301, cited at ~3732-3733). U -> V integral surjective gives V affine.

Source reference. Original proposition 07VT: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3641–3641. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03BZ: focused-seeds/geometry/S9-spaces-limits-integral-universally-bijective-scheme/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is integral and induces a bijection $|X| \to |Y|$. Then X is a scheme if and only if Y is a scheme.*

Proof. An integral morphism is representable by definition, hence if Y is a scheme, so is X. Conversely, assume that X is a scheme. Let $U \subset X$ be an affine open. An integral morphism is closed and $|f|$ is bijective, hence $|f|(|U|) \subset |Y|$ is open as the complement of $|f|(|X| \setminus |U|)$. Let $V \subset Y$ be the open subspace with $|V| = |f|(|U|)$, see Properties of Spaces, Lemma 03BZ. Then $U \to V$ is integral and surjective, hence V is an affine scheme by Proposition 07VT. This concludes the proof. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.